MILITARY SPECIFICATION

ELECTRONIC ASSEMBLY, FOR MINES,
M718A1/M741A1 SYSTEM

This Amendment forms a part of Military Specification MIL-E-48630 (AR), dated 31 July 1986, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Change "M718E1/M741E1" to "M718A1/M741A1".

PAGE 2

2.1.1, Under Specifications, Military: Delete "MIL-P-63238" and add "MIL-P-48649 - Projectile, 155mm, AT M718A1/M741Al Loading, Assembling and Packing of."

2.1.1, Under Drawing, Add new drawing:

"9317597 - Projectile, 155MM Test, Non-hazardous, M718 Type.

9317598 - Projectile, 155MM Test, Non-hazardous, M741 Type ."

PAGE 3

2.1.2:

Under DRAWINGS 9287096: Delete "Fixed Electrolytic." and substitute "Fixed Electrolyic, Tantalum Solid."

Under DRAWINGS: Add new drawing: 9287148 - ENCAPSULATION PROCEDURES AND REQUIREMENTS".

* Delete drawing "9342574" and "9342577" and add new drawing "11786216" and "11786241."

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PAGE 5

3.3.2.1: Change "M718E1/M741El" to "M718A1/M741Al".

* 3.4.1, last line: Delete “+125°F and substitute “+145°F”.

PAGE 6

* 3.5.1, Second line down:

Delete “300 microampere” and substitute “500 microampere.”

3.5.2.3, Last sentence: Delete “+125°F" and substitute “+145°F”.

3.5.2.4.1: Change last sentence from “A total of 392,448 primary time base pulses ...” to "A total of 1024 fast clock pulses and 391,424 primary time base pulses ..."

3.5.2.4.2: Change last sentence from "A total of 32000 primary time base pulses ...” to “A total of 1024 fast clock pulses and 30976 primary time base pulses ...".

PAGE 7

3.5.2.9: Delete the paragraph in its entirety.

3.5.5.2, Last sentence: Delete “DRSMC-QAT-M(D)” and substitute “AMSMC-QAT-M (D)”

3.5.7: Add new paragraph:

“3.5.7 Shorting bar open circuit. The resistance in the shorting bar loop when measured between the output terminal to ground shall be greater than 100K ohms (see 4.5.16.1).”

PAGE 8

* 3.7.1, third line down: Delete "sensor threshold level” and substitute “sensor fire angle.”

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PAGE 9

4.3.1:

"Delete Drawing “9287141" and substitute “9342554”.

Under DRAWING FOR PRINTED Wiring Board: Delete drawing “9287141" and substitute "9342554”.

PAGE 11

Table I. FIRST ARTICLE INSPECTION, under reference/inspection method:

Delete "4.4.2.27" and substitute “4.4.2.25”

Delete “4.4.2.28” and substitute “4.4.2.26”

Delete “4.4.2.25” and substitute “4.4.2.23”

Delete "4.4.2.26” and substitute “4.4.2.24”

PAGE 12

Table I. FIRST ARTICLE INSPECTION, under paragraph reference/ inspection method:

Delete “4.4.2.3” and substitute “4.4.2.7"

Delete “4.4.2.22" and substitute "4.4.2.26”

PAGE 13

* 4.4.1, subparagraph c: Delete in its entirety.

PAGE 15

4.4.2.2, Major 108: Delete quantity "10" under No. of sample units.

PAGE 16

4.4.2.3, Major 106: Delete “0.40%” under AQL or 100% and substitute “Note 2".

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PAGE 17

4.4.2.4, Major 101: Delete "Solderability (2 leads)" and substitute "Solderability".

PAGE 18

4.4.2.5:

Minor 201: Delete "3.12" under requirement paragraph and substitute "3.6".

* Delete the statement "missing category for plating."

4.4.2.5:

Minor 201: Under AQL, delete "1.0" and substitute "0.65".

PAGE 19

* 4.4.2.6: Delete "Note 1" in its entirety and substitute "(l) The AQL applies to the number of holes per board. If any board exceeds this criteria the lot shall be rejected. AQL is 0.40%".

PAGE 20

4.4.2.7:

Minor 201: Delete "None defined" under EXAMINATION OR TEST and substitute "Poor workmanship".

* Add new major:

"104 Location of terminal lug(s) incorrect 1/ 0.40% 3.2 Gage" .

PAGE 23

4.4.2.10, Under DRAWING NUMBER BLOCK: Delete drawing "9342560-x" and substitute "9342560".

PAGE 24

4.4.2.11: Delete major defect 105.

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PAGE 25

4.4.2.12: Delete major defect 105.

Major 107: Delete "Certification to note 3.4 of drawing" under Paragraph Reference/Inspection Method and substitute "Certification to Notes 3.4 and 3.5 of drawing".

PAGE 27

4.4.2.14: Delete major defect 105.

PAGE 28

4.4.2.15: Delete major defect 105.

PAGE 31

* 4.4.2.18, Under drawing number: Delete "RCR07GXXX/MIL-R-39008/" and substitute "RCR05GXXX/MIL-R-39008/".

PAGE 32

4.4.2.19, Major 102: Delete "MIL-R-55187-7" under paragraph reference/inspection method and substitute "MIL-R-55182/7".

PAGE 33

4.4.2.20, Major 103: Delete "3.5.3.2" under requirement paragraph and substitute "3.5.5.1".

PAGE 34

* 4.4.2.21: Delete the page in its entirety and substitute included page 34.

PAGE 35

* 4.4.2.22, Delete the page in its entirety and substitute included page 35.

PAGE 39

4.4.2.25, Major 104: Delete "3.5.2.2.4.1" under requirement paragraph and substitute "3.5.2.4.1".

* Major 104: Delete "4.5.7" under paragraph reference/inspection method and substitute "4.5.8".

* Special e: Delete "self-sterilization" and substitute "self-destruct".

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PAGE 40

4.4.2.25: Add the follwing Major defect:

Category	Exam/Test	No. of Sample Units	AQL	Reg. Para.	Para. Ref/ Insp. Method
* 126	Shorting Bar Open Circuit (NOTE 8)	32-0-1		3.5.7	4.5.16.1*

Add new note:

"(8) Use the units from the Real Time SD Timeout Test."

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.21	Transistor (JAN 2N222A)		MIL-S-19500/255
			NEXT HIGHER ASSEMBLY 9342553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bond wire not aluminum	32	32-0-1	3.2	Visual
102	MIL-S-19500/255 Compliance		3.2	3.2	Certification 1/
103	Solderability (3 leads ea.)	10	10-0-1	3.5.5.1	2/
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES:
1 Visual evidence of JAN marking.
2/ Use applicable inspection method per para 3.5.5.1

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.22	Transistor (JAN 2N2907A)	1	1	MIL-S-19500/291
				NEXT HIGHER ASSEMBLY: 9342553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bond wire not aluminum	32	32-0-1	3.2	Visual
102	MIL-S-19500/291 Compliance			3.2	Certification 1/
103	Solderability (3 leads ea.)	10	10-0-1	3.5.5.1	2/
Minor					
201	Poor workmanship				

NOTES:
1/ Visual evidence of JAN marking.
2/ Use applicable inspection method per para. 3.5.5.1

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PAGE 42

4.4.2.26:

Under drawing number: Delete "934255-2-1 & 2" and substitute "9342552-1 & 2".

Under requirement paragraph:

Critical 1: Delete "3.5.2.4.1" and substitute "3.5.2.4.2".

Critical 2: Delete "3.5.2.4.1" and substitute "3.5.2.4.2".

PAGE 43

4.4.2.26:
Major 104: Delete "3.5.2.2.4.1" under requirement paragraph and substitute "3.5.2.4.1".

* Delete "4.5.7" under paragraph reference/inspection method and substitute "4.5.9."

* Special e: Delete "self-sterilization" and substitute "self-destruct."

PAGE 44

4.4.2.26: Delete Major 123 and substitute the following major defect:

Category Major	Exam/Test	No.of Samples	Units	AQL	Req. Para.	Para Insp.	Ref/ Method
123	Real Time SD Timeout Low-Dud and early						
	Hot Temperature (Note 4)	8-1-3 16-2-3			3.5.2.4.1		4.5.26
	Cold Temperature (Note 5)	8-1-3 16-2-3			3.5.2.4.1		4.5.26
	Ambient (Note 3)	16			3.5.2.4.1		4.5.26

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4.4.2.26: Add the following Major Defect:

Category	Exam/Test	No. of Sample Units	AQL	Reg. Para.	Para. Ref/ Insp. Method
* 127	Shorting Bar Open Circuit (NOTE 8)	32-0-1		3.5.7	4.5.16.1*

Add new note:

"(8) Use the units from the Real Time SD Timeout Test."

PAGE 46

* 4.4.2.27: Add new paragraph included.

4.4.3.1, Second paragraph: Delete "4.4.2.27, 4.4.2.28" and substitute "4.4.2.25, 4.4.2.26".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.27	Form, Coil				9342556
					NEXT HIGHER ASSEMBLY 9342555
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	I.D. of mounting post(s) incorrect	1/	0.40%	3.2	Gage/4.4.5
102	O.D. of mounting post(s) incorrect	1/	0.40%	3.2	Gage/4.4.5
103	True position of mounting post hole(s) incorrect	1/	0.40%	3.2	Gage/4.4.5
104	Depth of mounting post hole(s) incorrect	1/	0.40%	3.2	Gage/4.4.5
105	Height of mounting post(s) incorrect	1/	0.40%	3.2	Gage/4.4.5
106	Width of ring incorrect (all around)		0.40%	3.2	Gage/4.4.5
107	Wall thickness incorrect (3 places)		0.40%	3.2	Gage/4.4.5
108	Height less mounting posts incorrect		0.40%	3.2	Gage/4.4.5
109	Profile of inside configuration incorrect		0.40%	3.2	Gage
110	Material		0.40%	3.2	Certificate/6.2
Minor					
201	Poor workmanship		0.65%	3.6	Visual

REMARKS

1/ Examination to be performed at 2 places.

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PAGE 47

4.4.5, Eleventh line down: Delete "4.4.2.28" and substitute "4.4.2.26".

4.5.6, Second line down: Delete "P/N 9342552-1" and substitute "P/N 9342551-1".

PAGE 50

4.5.12: Delete "3.5.2.7" and substitute "3.5.2.8".

4.5.16, Third line down: Delete "TP9" and substitute "TP2".

Add new paragraph:

"4.5.16.1 Shorting bar open circuit. The resistance of each shorting bar circuit shall be measured, using an approved ohmmeter, between the test points E3 and TP9. Observation shall be made for compliance with requirement of 3.5.7. This is a destructive test to be performed after the Real Time SD Timeout Test."

PAGE 51

4.5.18:

Paragraph title: Delete "old" and substitute "cold".

Fifth line down: Delete "tested atambient" and substitute "tested at ambient".

* Sixth line down: Delete "+3° -0°" and substitute "+5°F".

PAGE 52

4.5.24: Delete paragraph.

* 4.5.25: Delete drawing "9317597" and "9317598" and substitute "11786216" and "11786241" respectively.

* Delete "MIL-P-63238" in three places and substitute "MIL-P-48648".

Last sentence: Delete "MIL-P-63238" and substitute "MIL-P-48649".

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4.5.26 Sub paragraph d: Delete in its entirety and substitute:

"d. If an electronic assembly fails to function in the self-destruct mode after its maximum SD time period, the assembly shall be subjected to functioning by physical disturbance to promote anti-disturbance functioning, or applying a valid target signature using a simulated prime target sensor signature coil, or both. Any assembly failing to function (Non-Hazardous Dud) or function by disturbance or sensor input after SD time (Hazardous) shall be considered defective and the battery voltage shall then be measured and recorded and electronic assembly set aside for dismantling and fail analysis.

6.1: Change "M718/M741" to M718A1/M741Al".

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